# ZAKI IRSYADA PUTRA

**Software & AI Systems Engineer | Information Systems Undergraduate | National Business Case Champion**

Surabaya, Indonesia • **+62 821-9806-9550** • **irsyadazaki77@gmail.com**

**github.com/irsyadazaki77-beep** • **linkedin.com/in/zaki-irsyada-putra-23870a29b** • **irsyadazaki77-beep.github.io/portfolio**

## PROFESSIONAL SUMMARY

Information Systems undergraduate at Telkom University Surabaya (2025–Present) and **1st Place National Business Case Champion (INSIGHT 2025)** specializing in **Full-Stack Software Engineering**, **Applied Generative AI**, and **Enterprise Data Systems**. Proven builder of production-grade TypeScript, React 19, Node.js, PostgreSQL, Three.js/WebGL, WebRTC, and Google Gemini systems backed by automated Vitest/Playwright verification, alongside B2B commercial operations leadership and Komdigi/Microsoft certifications in Data Science and AI Security.

## WORK & ORGANIZATIONAL EXPERIENCE

**Marketing Manager** — **PT Agra Danapati Parama** 2025 – 2026

*Tourism & Travel Industry | Bekasi, Indonesia*

- Led B2B sales outreach, pitching corporate travel and tourism solutions to prospective enterprise clients.
- Designed and executed multi-channel outbound acquisition workflows to expand corporate client pipeline and product visibility.
- Administered a centralized **marketing & CRM database** covering lead segmentation, interaction tracking, and campaign analysis.
- Coordinated cross-functional sales-team execution to maintain pipeline velocity, target alignment, and management reporting.

**Organizational Leadership** — **Ikatan Pelajar Muhammadiyah (IPM) — SMA Muhammadiyah 1 Gresik** 2022 – 2025

*Gresik, Indonesia*

- **Head of Logistics & Security Division (2022 – 2025):** Planned, controlled, and audited divisional operational & procurement budgets while leading on-ground security and risk mitigation across major school events.
- **Member, Science & Knowledge Development / PIP (2022 – 2024):** Managed resource budgets for annual Bulan Bahasa programs, launched the weekly Pojok Literasi initiative, and supported event safety SOP execution.

## FEATURED ENGINEERING & AI PROJECTS

**RuangLABmu — 3D Virtual STEM Lab & Real-Time Physics Engine** Nov 2026 | **GitHub**

*TypeScript, React, Three.js / WebGL (EffectComposer, Bloom, FXAA), Web Workers, WebSockets, Gemini AI, Node Test*

- Engineered **28 interactive STEM modules** across 4 domains (Physics 10, Chemistry 6, Biology/3D Anatomy 7, Astronomy 5) on a custom Three.js/WebGL studio rendering pipeline with WebGL2 MSAA fallback.
- Built a **60Hz RK4 numerical solver** inside Web Workers with Gaussian instrument noise, paired with /ws-lab collaborative PIN rooms (20Hz 3D spatial cursors, mutex parameter locks, teacher freeze, exam mode, and replay).
- Integrated **Dr. LabMu** Socratic & Vision AI for automated experimental error detection ($R^2$ < 0.85 anomalies); verified with **50/50 automated tests passing** across 3 suites.

**RuangTenang Kampus v4.0 — AI Mental Wellness & RuangKerja Studio** Sep – Nov 2026 | **GitHub · Live Demo**

*React 19, TypeScript, Express 4, Prisma 6, PostgreSQL 16 / SQLite, Redis 7, Gemini / OpenAI / DeepSeek / Ollama, Docker, Nginx*

- Architected a full-stack student wellness ecosystem featuring **multi-provider CBT AI** (cloud LLMs + Ollama zero-egress offline) with SSE streaming, regex PII scrubbing, and emergency SOS crisis triage.
- Built **RuangKerja Artifact Studio** (Monaco Editor, Tiptap, Mermaid, KaTeX, DOCX/PPTX/PDF/XLSX export + Voice Defense Simulator) alongside validated PHQ-9 and GAD-7 clinical screenings.
- Enforced **AES-256-GCM envelope encryption**, multi-tenant k-anonymity analytics guards, and ACID-safe multi-timezone (WIB/WITA/WIT) counselor scheduling preventing double-booking.

**RuangPintar Edge — Offline-First WebRTC P2P Mesh & Edge AI LMS** Nov 2026 | **GitHub**

*React, TypeScript, Node.js, WebRTC (RTCDataChannel), SQLite WAL, IndexedDB PWA, Google Gemini 3.8 Flash, AES-GCM*

- Built a decentralized LMS for 3T schools distributing curriculum bundles in 16 binary chunks (32-byte RPE1 frame) over **WebRTC SCTP RTCDataChannel** at **0 KB ISP data cost**.
- Engineered a Single-Gateway Teacher edge hub querying a local **SQLite WAL semantic cache** (≥0.45 similarity, 0 ms repeat hits) before falling back to gemini-3.8-flash.
- Implemented IndexedDB offline persistence, **AES-GCM encrypted .json Sneakernet USB sync**, and 6 interactive STEM simulations resilient to intermittent school connectivity.

**ZiGame 2.0 — Full-Stack Web Arcade & Server-Authoritative Economy** Nov 2026 | **GitHub**

*React 19, TypeScript, Three.js, Vite, Express Serverless, Cloud Firestore, PWA, Vitest, Playwright*

- Developed **39 playable 2D/3D browser mini-games** on a standardized **60 FPS useGameEngine SDK** with offline PWA score queueing and instant restart loops.
- Implemented **server-authoritative anti-cheat** (/api/session/start signed nonces, server duration & score bounds), double-entry coin ledger, X-Idempotency-Key shop guards, and Elo/MMR leaderboards.

**ImpactGrid — Autonomous RAN Energy & ESG Decision Support System (v1.0.0-rc1)** Oct 2026 | GitHub

*React, TypeScript, O-RAN RIC rApp Simulation, 0-1 Knapsack, Leaflet GIS, Vitest, Playwright*

- Architected an **O-RAN Non-RT RIC rApp** optimization DSS targeting Telkomsel's **1.885B kWh/yr** (1.72M tCO2e/yr Scope 2) footprint with >75% PRB SLA protection and a **<3s Emergency 112 Kill-Switch**.
- Built a **Bayesian 95% CI Digital Twin** (40 pilot + 10 control sites), 38-province GIS, SAP S/4HANA IFRS 16 journals, SHA-256 BA-RE legal PDFs, and GRI 302/305 audit chain modeling Rp95M G0 / Rp450M pilot ROI up to **+Rp41.48B projected 5-yr NPV** (14.60 GWh/yr); **62/62 Vitest tests passing** + Playwright E2E.

**SiKaya — Multi-Asset Investment Education & AI Advisory Platform** Nov 2026 | GitHub · Live Demo

*React 19, TypeScript, Express Serverless, Firebase Firestore, Gemini AI, Helmet, Zod, Vitest, Supertest, Playwright*

- Engineered a multi-asset simulator (equities, crypto, bonds, gold) on Vercel Serverless using a 30s memory cache, request coalescing, and 5m SWR market fallback plus **Gemini Voice & Text Advisory**.
- Hardened data and API boundaries with **anti-tamper Firestore Security Rules**, Helmet, 3-tier rate limiting, Zod validation, and X-Request-ID tracing; verified with **25/25 Vitest/Supertest API/unit tests passing** alongside Playwright E2E & axe-core WCAG suites.

**Additional Engineering Systems & Prototypes (Sep – Nov 2026)**

- **VoxelVerse 1.0** (GitHub · Live): Procedural 3D voxel survival sandbox using React, Three.js, Web Worker greedy meshing, 17 biomes, ws authoritative multiplayer, and a Tauri (Rust) native Windows desktop bundle.
- **OmniTRAF Surabaya** (GitHub): Real-time traffic command center integrating Node.js/Express, Socket.io 300ms telemetry, Leaflet vector GIS, 60 FPS Canvas Edge Vision HUD, IPCC 2006 carbon model, and WCAG 2.1 AA PWA support.
- **Desktop & UI/UX Systems**: Dynamic Island Pro (.NET 9 / WPF, Hooke's Law VSync physics & Windows GSMTC live lyrics); QuickLook for Windows (C# / .NET 9, Space-key instant file previewer); KITORANG (INSIGHT 2026 Central Papua Fair-Trade Ethno-Commerce UI/UX prototype).

## HONORS & COMPETITIONS

- **1st Place (Juara 1)** — Business Case Competition INSIGHT 2025 — HMSI Telkom University Surabaya **Sep 2025**
- **Semifinalist** — BCC TDC Summit Fest 2026 (Telkomsel ESG Case, Team Leader, Score 77.5) — ITS Surabaya **2026**
- **Semifinalist** — Business Model Canvas DIGITOPIA FAIR 4.0 — Telkom University Surabaya **Jun 2026**
- **Semifinalist** — YED UI Business Plan Competition (National University Category) — BEM FMIPA UI **Nov – Dec 2025**
- **Semifinalist** — IMAGEPRENEUR 2025 (Team Kacamata Usang, Tech for Change) — UPI Bandung **Dec 2025**

## EDUCATION

**Telkom University Surabaya** — Bachelor of Information Systems (S1 Sistem Informasi) **2025 – Present**

Focus: Full-Stack Software Engineering, Applied AI Systems, Distributed Architectures, and Enterprise Data Analytics.

**SMA Muhammadiyah 1 Gresik** — High School Diploma, Natural Sciences (MIPA) **2022 – 2025**

Active in Ikatan Pelajar Muhammadiyah (IPM) Logistics & Security leadership and Science & Knowledge (PIP) programs.

## CERTIFICATIONS & TRAINING

- **Associate Data Scientist + Python (31 Hrs)** & **Data Scientist Supervisor (20 Hrs)** — Digital Talent Academy, Komdigi (Apr – May 2026)
- **Fundamental of Microsoft Fabric & Azure (17 Hrs)** & **Fundamentals of AI Security** — Komdigi & Microsoft (Dec 2025 – May 2026)
- **Belajar Dasar AI / Basic Artificial Intelligence** (ID: GRX5JWY9KX0M) — Dicoding Indonesia (Nov 2025 – Nov 2028) | Applied Programming — Prodistik ITS (2025)
- **Excel Data Analysis** & **Project Management Fundamentals** — Microsoft Skills for Jobs (Dec 2025)
- **Komdigi DTS Micro-Credentials (8 Modules):** AI Engineering, AI Implementation, Cybersecurity, Ethical Hacking, Personal Data Protection (Nov – Dec 2025)

## TECHNICAL SKILLS

**Languages:** TypeScript, JavaScript (ES6+), Python, C# (.NET 9), SQL, HTML5, CSS3

**Frontend & 3D Graphics:** React 19, Vite, Tailwind CSS, Three.js / WebGL (EffectComposer, GLSL), Web Workers, Leaflet GIS, PWA

**Backend, Real-Time & DB:** Node.js, Express.js, Prisma ORM, PostgreSQL 16, SQLite (WAL), Redis 7, Firebase Firestore, Socket.io, WebRTC

**AI, Security & Desktop:** Google Gemini API (@google/genai), Multi-LLM / Ollama Integration, AES-256-GCM, Helmet, Zod, Firestore Rules, Tauri (Rust), WPF

**Testing, Cloud & Analytics:** Vitest, Playwright E2E, Supertest, Docker, Nginx, Git/GitHub, Vercel, Microsoft Excel, Microsoft Fabric & Azure